How the Computer Communicates - Binary

Watch the video: https://www.youtube.com/watch?v=u4P0LOofEFs

Computers use a number system called “Binary” to process information. Binary is a number system based on 0’s and 1’s. The reason for binary is that it is easily converted to an electrical state: 0 = off, 1 = on.

Decimal : We use the Decimal system
(Base _10_ number system – 10 digits – 0, 1, 2,… 9)
Reason we use decimal – 10 fingers

2495=two thousand four hundred and ninety five

In binary this would be 1001 1011 1111

Counting in Decimal : 0 , 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11,...

Binary : Counting in Binary :
0,1,10,11,100,101,110,111,1000,1001,1010,1011...

Units of Information Measurement

Data is stored on a computer using bits.
With 1 bit you can store 2 pieces of data.
With 2 bits you can store 4 pieces of data.
With 3 bits you can store 8 pieces of data.
and so on……!

2 exponent number of bits=total number of representations
2^3=8
2^4=16
Normally bits are grouped together into groups of 4. This is called a ___nibble_____. A group of bits is called a ____byte___.

Computers normally use these as their standard unit of measuring data/memory. Units are used to measure all kinds of things i.e.

__meters___ for measuring distance, gram for mass, litre for volume. Computers use the byte for measuring data.

Byte Units

Unit	Equivalent
1 kilobyte (KB)	1,024 bytes
1 megabyte (MB)	1,048,576 bytes
1 gigabyte (GB)	1,073,741,824 bytes
1 terabyte (TB)	1,099,511,627,776 bytes
1 petabyte (PB)	1,125,899,906,842,624 bytes

We can approximate these by using factors of 1000 instead of 1024. This means there are approximately 1000 bytes in a KB and 1000 KB in a MB and 1000MB in a GB and so.

Example Problems: Convert the following into their approximate equivalent values.
4200B=0.0042 MB
1900KB=1,900,000 B
50MB=50,000,000 B
2.5TB=2500 GB

Convert the following into their exact equivalent values.
4200B=4200/1024/1024=0.004MB
1900KB=1900x1024=1,945,600B
50MB=50x1024x1024=52,428,800B
2.5TB=2.5x1024=2560GB

Groups of 4 bits are normally written with a space between them i.e. 1100 0011.

A group of 4 bits is called a ___Nibble_____.

Even 1 bit can be written as a nibble. For example the binary number 1 can be written as __0001_____. Any left trailing zeroes do not affect the value of any number in any number system.

Examples: Write the following binary numbers in groups of nibbles:
1=__0001_____
010=_0010_____
1101101=__0110 1101_____
11=_0011_____
010101010101=__0101 0101 0101_____

Since computers only understand binary and we only understand decimal, engineers and mathematicians had to find a way to convert between them so that we could use computers easily.

Exercises-using the internet and these notes, find answers to the following questions:
1. Find and list 3 websites and 3 youtube videos that discuss binary numbers.
https://www.w3schools.com/programming/prog_binary_numbers.php
https://www.purplemath.com/modules/numbbase.htm
https://www.electronics-tutorials.ws/binary/bin_1.html
https://www.youtube.com/watch?v=FrsgHUSZ_pc
https://www.youtube.com/watch?v=kTcpd4ef2IU
https://www.youtube.com/watch?v=ImJSldHR0WA
2. Do some research and find out what are some ways we use to represent binary numbers.
digital signals, such as high/low voltages or on/off states, that correspond to the binary digits (bits) 1 and 0.

3. What base system is binary?
Base 2
4. What is a single binary digit called?
A bit
5. What is a nibble?
4 bits
6. What is a byte?
8 bits
7. How many things can I represent with 1 bit?
2
8. How many things can I represent with 2 bits? 4 bits? 8 bits?
2 bits: 4
4 bits: 8
8 bits: 16

9. Write down all the combinations of 3 bits.

000
001
010
011
100
101
110
111

10. There are many online tools or sites that will help you convert decimal to binary. Find and use these to fill in the following table:

Decimal	Binary
20	0001 0100
31	0001 1111
32	0010 0000
63	0011 1111
14	1110
51	0011 0011
68	0100 0100
71	0100 0111
18	0001 0010
13	1101
5	0101
163	1010 0011
255	1111 1111
95	0101 1111
113	0111 0001
91	0101 1011
15	1111

Convert the following into their approximate and exact equivalent values:

32,000 MB=
32000000000 B

4,000
GB=4000000 MB

6,000 B=
0.000000006TB

75,000 TB=
75000000GB

26,000 KB=
0.000000026PB

2,000,000 MB=2 TB

64,000,000 B=
64000KB

32,000,000 MB=
32000000000KB

8,000,000 GB=
8000000000 MB

16,000,000 PB=
160000000TB

11. Convert 8 bit into Byte (highlight one).

1 Byte
2 Byte
8 Byte
4 Byte

12. Convert 2GB into kB(highlight one).

2x1024x1024 kB
2x1024x1024x1024 kB
2/1024/1024 kB

2/1024 kB

13. Byte is (highlight one)

Smallest unit of data computer can process

A group of 8 bits

14. Choose 3 answers below (highlight three) :

20 MB is equal to

20480 kB

20971520 B

20/1024/1024TB

20TB

15. Pick 2 options (highlight two).

1 TB =1024GB

1024 GB=1 TB

1024x1024 MB

1024 /1024 MB

16. Choose 3 answers below (highlight three) :

20 MB is equal to

20480 kB

20971520 B

20/1024/1024TB

20TB